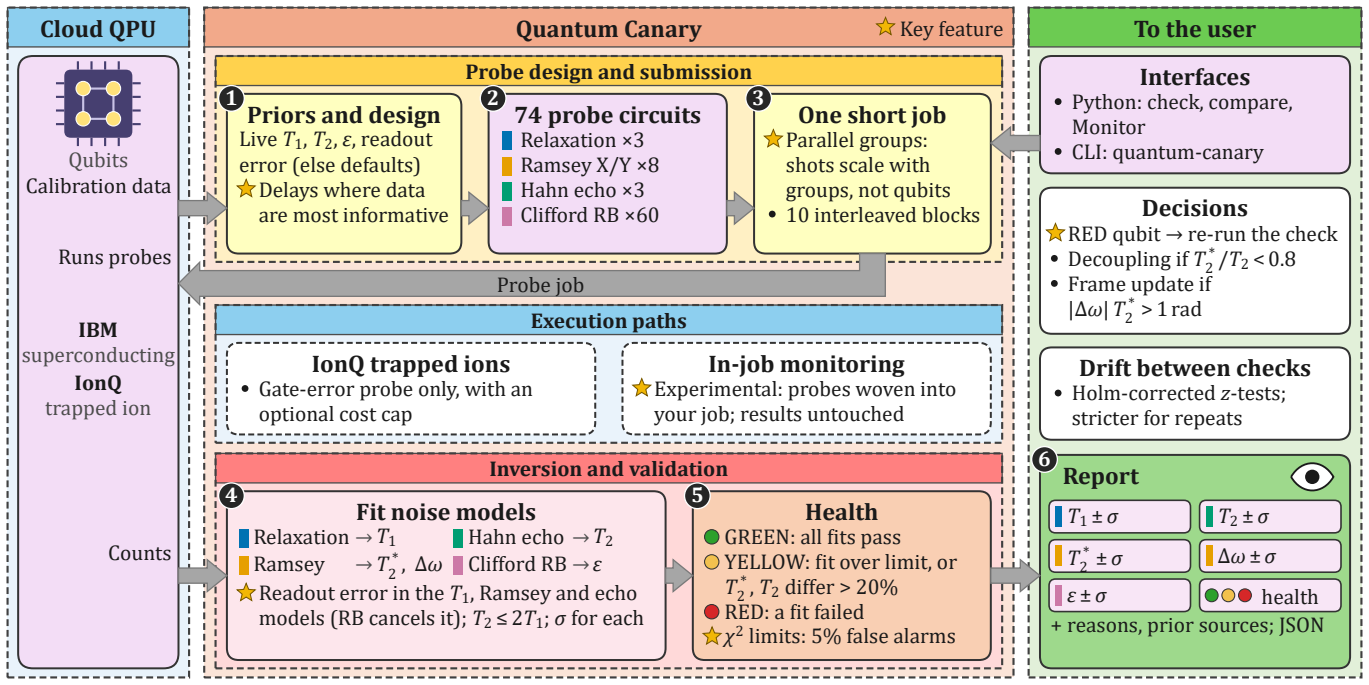

Cloud QPU
Quantum Canary
Key feature
To the user
Qubits
Calibration data
Runs probes
IBM
superconducting
IonQ
trapped ion
Counts
Probe design and submission
1
Priors and design
Live T_1, T_2, ε, readout error (else defaults)
Delays where data are most informative
2
74 probe circuits
Relaxation ×3
Ramsey X/Y ×8
Hahn echo ×3
Clifford RB ×60
3
One short job
Parallel groups: shots scale with groups, not qubits
10 interleaved blocks
Probe job
Execution paths
IonQ trapped ions
Gate-error probe only, with an optional cost cap
In-job monitoring
Experimental: probes woven into your job; results untouched
Inversion and validation
4
Fit noise models
Relaxation → T_1
Hahn echo → T_2
Ramsey → T_2^*, $\Delta\omega$
Clifford RB → ε
Readout error in the T_1, Ramsey and echo models (RB cancels it); $T_2 \le 2T_1$; σ for each
5
Health
GREEN: all fits pass
YELLOW: fit over limit, or T_2^*, T_2 differ > 20%
RED: a fit failed
χ^2 limits: 5% false alarms
Interfaces
Python: check, compare, Monitor
CLI: quantum-canary
Decisions
RED qubit → re-run the check
Decoupling if $T_2^*/T_2 < 0.8$
Frame update if $|\Delta\omega|\,T_2^* > 1$ rad
Drift between checks
Holm-corrected z-tests; stricter for repeats
6
Report
$T_1 \pm \sigma$
$T_2 \pm \sigma$
$T_2^* \pm \sigma$
$\Delta\omega \pm \sigma$
$\varepsilon \pm \sigma$
health
+ reasons, prior sources; JSON